

Problem 15.26

For \$10,000, Kelly purchases an annuity-immediate that pays \$400 quarterly for the next 10 years. Calculate the annual nominal interest rate convertible monthly earned by Kelly's investment. Hint: Use linear interpolation.

Solution

Since payments are made quarterly for 10 years, the number of payments is

$$n = 10(4) = 40.$$

Let j be the effective quarterly interest rate. The purchase price is the present value of the quarterly payments, so

$$10000 = 400a_{\overline{40}|j}.$$

Thus,

$$25 = a_{\overline{40}|j}.$$

Using

$$a_{\overline{n}|j} = \frac{1 - (1 + j)^{-n}}{j},$$

we need

$$25 = \frac{1 - (1 + j)^{-40}}{j}.$$

Using linear interpolation, compare the annuity values near 25:

$$\frac{1 - (1.023)^{-40}}{0.023} \approx 25.96$$

and

$$\frac{1 - (1.02532)^{-40}}{0.02532} \approx 24.968.$$

Interpolating to $a_{\overline{40}|j} = 25$ gives

$$\begin{aligned} j &\approx 0.023 + (0.02532 - 0.023) \left(\frac{25 - 25.96}{24.968 - 25.96} \right) \\ &\approx 0.02524. \end{aligned}$$

Now convert this quarterly effective rate to an annual nominal rate convertible monthly. The equivalent annual accumulation factors must satisfy

$$(1 + j)^4 = \left(1 + \frac{i^{(12)}}{12} \right)^{12}.$$

Taking cube roots gives

$$1 + \frac{i^{(12)}}{12} = (1 + j)^{1/3}.$$

Therefore,

$$i^{(12)} = 12 \left((1 + j)^{1/3} - 1 \right).$$

Substituting $j \approx 0.02524$,

$$\begin{aligned} i^{(12)} &\approx 12 \left((1.02524)^{1/3} - 1 \right) \\ &\approx 0.10014. \end{aligned}$$

Hence, the annual nominal interest rate convertible monthly is

$$\boxed{i^{(12)} \approx 10.01\%}.$$